

RUO'ERGA, China

WMO: 560790

Lat: 33.5748N

Lon: 102.9660E

Elev: 3441

StdP: 66.26

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-20.8	-18.5	-27.3	0.5	-10.7	-25.5	0.6	-10.3	9.7	-1.2	8.4	-0.4	0.9	40	0.701

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.5	20.9	12.2	19.5	11.6	18.1	10.9	13.5	19.1	12.7	17.9	12.0	16.6	2.5	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
11.4	12.9	16.2	10.7	12.3	15.3	10.1	11.8	14.5	50.8	19.3	48.3	18.0	45.8	16.7	16.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%		2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
Min		Max		Min	Max		Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)		(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8.2	6.9	5.9		-24.9	23.6	2.1	1.1	-26.4	24.4	-27.6	25.0	-28.8	25.6	-30.3	26.4
				-25.2	14.9	2.0	0.7	-26.7	15.4	-27.9	15.8	-29.0	16.2	-30.5	16.8
				DB											
				WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	2.9	-7.9	-5.1	-0.7	3.6	6.7	9.8	12.2	11.8	8.8	3.7	-2.0	-6.7	(6)
(7)		DBStd	7.56	3.47	3.78	3.37	3.13	2.62	2.49	2.38	2.57	3.03	2.90	3.34	3.18	(7)
(8)		HDD10.0	2793	554	424	332	194	106	33	8	11	59	197	359	517	(8)
(9)		HDD18.3	5637	813	657	591	443	360	256	192	201	286	454	609	776	(9)
(10)		CDD10.0	199	0	0	0	1	4	27	74	68	23	1	0	0	(10)
(11)		CDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0	(11)
(12)		CDH23.3	2	0	0	0	0	0	0	1	1	0	0	0	0	(12)
(13)		CDH26.7	0	0	0	0	0	0	0	0	0	0	0	0	0	(13)
(14)	Wind	WSAvg	2.4	2.3	2.6	2.8	2.8	2.7	2.4	2.2	2.1	2.3	2.2	2.0	2.0	(14)
(15)	Precipitation	PrecAvg	661	4	7	17	37	79	97	126	112	116	56	8	2	(15)
(16)		PrecMax	863	11	20	27	75	118	192	226	199	214	144	20	7	(16)
(17)		PrecMin	494	0	0	7	15	49	60	61	54	50	9	0	0	(17)
(18)		PrecStd	100	3	5	5	12	20	32	37	39	47	33	5	2	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	8.2	11.2	13.6	17.5	19.1	20.5	22.5	22.7	21.2	17.1	11.3	8.5	(19)
MCWB			-1.6	0.0	2.1	5.3	8.4	11.7	13.1	13.2	11.9	8.7	1.4	-0.6	(20)	
2%		DB	5.2	8.0	10.7	14.5	16.6	18.2	20.8	20.8	18.8	13.6	8.9	5.4	(21)	
		MCWB	-2.9	-1.5	0.9	4.2	7.3	10.6	12.4	12.5	10.9	5.9	0.5	-2.4	(22)	
5%		DB	2.9	5.7	8.4	12.5	14.8	16.7	19.4	19.2	16.6	11.4	7.1	3.3	(23)	
		MCWB	-4.1	-2.3	0.0	3.3	6.5	9.8	11.9	11.8	10.0	4.7	-0.3	-3.6	(24)	
10%		DB	0.7	3.4	6.2	10.5	13.0	15.2	17.9	17.7	14.5	9.5	5.3	1.3	(25)	
		MCWB	-5.2	-3.3	-0.9	2.8	5.9	9.1	11.4	11.1	8.8	3.8	-1.1	-4.5	(26)	
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-1.1	0.5	3.1	6.7	10.4	13.5	14.5	14.3	13.3	9.8	2.6	-0.2	(27)
MCDWB			7.3	10.0	11.5	14.3	16.3	18.6	20.4	20.7	19.0	16.3	9.2	7.8	(28)	
2%		WB	-2.6	-1.0	1.7	5.3	8.8	12.2	13.5	13.4	11.8	7.5	1.4	-2.0	(29)	
		MCDWB	4.9	6.5	8.7	12.2	14.2	16.3	19.0	19.3	17.3	11.7	7.3	4.7	(30)	
5%		WB	-3.8	-2.0	0.6	4.3	7.7	11.1	12.8	12.6	10.8	5.9	0.2	-3.3	(31)	
		MCDWB	2.5	4.9	7.0	10.5	12.9	15.0	17.9	18.0	15.5	9.8	5.6	2.6	(32)	
10%		WB	-5.0	-3.0	-0.4	3.3	6.6	10.1	12.0	11.8	9.7	4.6	-0.7	-4.3	(33)	
		MCDWB	0.4	2.7	5.0	9.0	11.5	13.7	16.4	16.5	13.4	8.1	4.2	0.9	(34)	
(35)	Mean Daily Temperature Range	5% DB	MDBR	16.3	15.2	12.1	11.7	11.0	9.7	10.5	10.8	10.3	10.5	13.4	15.6	(35)
MCDBR			20.0	19.5	16.1	15.9	14.7	12.5	13.5	13.4	13.6	14.0	16.0	18.5	(36)	
MCWBR			12.9	11.4	7.7	6.8	6.6	5.8	6.1	6.1	6.4	6.9	8.4	11.6	(37)	
5% WB		MCDBR	18.5	17.6	13.9	13.3	12.0	9.5	10.9	11.1	11.3	11.2	14.0	17.1	(38)	
		MCWBR	12.1	10.4	6.9	6.3	6.1	4.8	5.3	5.2	5.5	6.1	7.7	10.8	(39)	
(40)	Clear-Sky Solar Irradiance	taub	0.242	0.274	0.333	0.353	0.345	0.337	0.337	0.338	0.312	0.276	0.243	0.223	(40)	
(41)		taud	2.471	2.383	2.200	2.196	2.311	2.412	2.434	2.419	2.488	2.546	2.550	2.558	(41)	
(42)		Ebn at Noon	993	984	940	932	939	941	939	936	954	972	987	1006	(42)	
(43)		Edh at Noon	85	104	137	144	130	118	114	114	102	88	78	73	(43)	
(44)	All-Sky Solar Radiation	RadAvg	3.43	4.08	4.88	5.42	5.43	5.08	5.54	5.19	4.21	3.81	3.57	3.26	(44)	
(45)		RadStd	0.17	0.24	0.23	0.27	0.24	0.46	0.57	0.57	0.35	0.38	0.13	0.14	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	+0.72	+1.37	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (6 neighbors)	+0.35	N/A	N/A	N/A	N/A	N/A	-43	-95	+56	N/A

Nomenclature: See separate page